

■ Burnout Rate AI Engine – Technical Blueprint

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■ 1. Input Data Required

A. Self-reported inputs:

- Mood (scale or emoji)
- Perceived stress (scale 1–10)
- Sleep quality (hours, restfulness)
- Energy level (low/medium/high)
- Workload (hours, task count)
- Emotional fatigue (e.g., “Do you feel emotionally drained?”)
- Social interaction level

B. Wearable/sensor data:

- HRV (Heart Rate Variability)
- Resting heart rate
- Sleep stages and total duration
- Steps/activity level
- Skin temp/GSR (if available)



■ 2. Burnout Score Formula (0–100)

Burnout Score = $w_1 \cdot \text{MoodScore} + w_2 \cdot \text{Stress} + w_3 \cdot \text{SleepDebt} + w_4 \cdot (1/\text{HRV}) + w_5 \cdot \text{Workload} + w_6 \cdot \text{EmotionalFatigue} + w_7 \cdot \text{RecoveryBuffer}$

- w_1 – w_7 are AI-learned weights or expert-defined multipliers.



■ 3. AI Approaches to Estimate Burnout Rate

- Rule-based scoring: Maslach Burnout Inventory (MBI)
- ML Regression Model: scikit-learn, XGBoost, LightGBM
- Time-series model: Prophet, LSTM
- Bayesian model: PyMC or Stan
- LLM + Prompting: Sentiment/emotion inference from journal entries



■ 4. Output Sample (Console UI)

Burnout Score: 72/100 ■ High Risk
Stage: Early Stage Burnout – Emotional Exhaustion
Insight: HRV dropped 22% over 3 days. Emotional fatigue rising.

Recommendation: Reduce workload by 20%. Take a mindful break.



■ 5. Visualization Options

- Trendline: Burnout score over 7–14 days
- Risk zones: Green (Low), Yellow (Moderate), Red (High)
- Daily insights: With emojis, icons, or charts
- Planner Suggestions: Yoga, Break, Music, Breathwork



■ Backend Integration

- Module: burnout_engine.py
- Trigger: After daily check-in + wearable sync
- Output: JSON, DB write, and UI display